

RESEARCH ARTICLE

Storm-Induced Solute Dynamics and Hysteresis Behaviour in an Eogenetic Karst Aquifer

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ABSTRACT

Eogenetic karst aquifers that maintain high carbonate bedrock permeability can have distinctive aquifer hydrodynamics that would be captured in hysteresis behaviour. Analysing the hysteresis behaviour of these systems can be a concise and efficient method to provide insight into the dominant recharge mechanisms, porosity integration and contributing land uses to spring discharge. The availability of deployable water quality sensors that can monitor spring water chemistry at high temporal resolution across multiple events allows us to capture delayed or attenuated signals that may occur in eogenetic karst. Additionally, analysing hysteresis across multiple events can identify patterns in the responses that occur. The Upper Floridan Aquifer (UFA) is an example of an eogenetic karst aquifer that has extensive phreatic conduits embedded in a high permeability carbonate matrix. We collected high temporal resolution (15 min) discharge, specific conductance and nitrate. We analysed storm-induced changes to nitrate and specific conductance and quantified hysteresis by calculating the hysteresis index (HI) and complementary flushing index (FI) at two major springs that drain agriculturally dense regions of the UFA. The combined analysis of HI and FI showed that 95% of all events had delayed connectivity of discrete feature recharge emerging at each spring. Mobilisation ($FI > 0$) of specific conductance occurred 95% of the time at both springs, but nitrate was diluted ($FI < 0$) 85% of the time at one spring and mobilised 95% of the time at the other. Changes to both specific conductance and nitrate were less than 15% of the pre-storm values, which illustrates how discrete recharged water mixes with substantial volumes of older, stored water in the aquifer. The hysteresis behaviour in the UFA was in contrast with a telogenetic setting, where higher variability in nitrate responses occurred between events, dilution was mostly observed for specific conductance and nitrate, and responses varied between delayed and rapid connectivity. Our work quantifies the storm-induced hydrodynamics of an eogenetic karst aquifer, which can help guide local water resource management decisions and advance our knowledge of karst aquifer hydrodynamics across a wide range of diagenetic and karstification stages.

1 | Introduction

Karst aquifers are among the most productive and widely used aquifers globally, representing 25% of the terrestrial landscape and supporting up to 10% of the worldwide population with

potable and irrigable water (Kuniansky et al. 2023). Karst aquifer productivity results from their characteristic solutional porosity features such as integrated conduits, fractures and voids that can quickly store and transport large volumes of water into and through these aquifer systems (Worthington 2007;

White 2018). The global benefit of karst aquifers is offset as the same solutional features that make them productive also make them highly susceptible to contamination. Local variations in landscape hydrology, climate, diagenetic maturity and the evolution of secondary porosity create unique flowpath distributions across individual karst aquifers (Worthington et al. 2000). Therefore, attempts to characterise karst aquifer flow dynamics are ongoing as local efforts are often needed for targeted water resource management (Goldscheider 2019). However, this site-specific knowledge ultimately leads to furthering our understanding of the variability in karst aquifer hydrology across a wide range of diagenetic maturity and karstification, ultimately leading to advancements in water resource management.

One way to characterise karst aquifer flow dynamics is by analysing event scale hysteresis patterns at karst springs. Hysteresis effects in karst systems are driven by recharge mechanisms, conduit network integration, conduit-matrix interactions and temporal changes in storage and transport processes (Jourde et al. 2007; Hartmann et al. 2014). These effects can be observed in high-frequency discharge–concentration (Q – C) relationships, which can be used as a diagnostic framework for identifying the contributions of different flow components (e.g., quickflow from conduits vs. slow flow from the matrix) to spring discharge. Though local interpretation is necessary (Liu et al. 2021), hysteresis behaviour characterises the efficiency of solute transport from recharge by describing Q – C relationships as clockwise (CW) or counterclockwise (CCW) patterns (Seeger et al. 2004; Koenig et al. 2017; Fang et al. 2008; Pellegrini et al. 2023). Hysteresis analysis can be strengthened by analysing the mobilisation and dilution of solutes that impact the hysteresis patterns (Waite et al. 2023), thereby helping to identify the land use types contributing to aquifer recharge (Lloyd et al. 2016; Husic et al. 2023). For example, CW hysteresis under solute mobilisation can indicate a rapid connection with surface contamination, while solute mobilisation with CCW hysteresis illustrates a delayed connection with the surface source that can be related to porosity integration and hydraulic efficiency (Zarnaghsh and Husic 2021; Husic et al. 2023). Furthermore, hysteresis complexity (e.g., Figure-8 curves) can provide information about changing solute sources and hydraulic conditions during a single event as well as identify potential solute release from storage (Williams 1989; Lloyd et al. 2016; Bettel et al. 2022). Hysteresis analysis has broad, informative applications and has been used to characterise the vulnerability of karst aquifers to contamination events, understand the landscape elements contributing to recharge, identify conduit porosity integration and improve solute transport modelling and land use planning in contaminant-prone karst aquifers (Massei et al. 2007; Mahler and Massei 2007; Opsahl et al. 2017; Schilling et al. 2019; Zhang et al. 2020; Sun et al. 2023; Husic et al. 2023). However, studies of Q – C behaviour in karst have largely occurred in classic, telogenetic karst where low matrix permeability focuses flow along discrete features (Moore et al. 2010), and thus more representative studies in aquifers characterised by carbonate matrix control remain (Spellman, Gulley, et al. 2022).

Eogenetic karst is still in the early diagenetic stage and therefore typically retains primary porosity and high matrix permeability (Choquette and Pray 1970; Vacher and Mylroie 2002).

Furthermore, porosity enhancement of vugs and fractures via diffuse recharge to the water table enlarges over time and coalesces, creating caves and eventually forming distributed cave systems that may be disconnected or integrated across the aquifer (Florea 2006a, 2006b; Budd and Vacher 2004; Kaufmann 2016; deRoij and Graham 2017). In contrast, telogenetic karst develops when carbonate rocks have undergone significant burial and lithification, leading to the loss of primary porosity. These rocks have been uplifted and exposed to surface weathering processes, and caves form through focused autogenic and allogenic recharge, which generates the large, integrated expansive conduit networks found in most classical karst (Palmer 1991; Ford and Williams 2007). As such, comparisons of spring hydrographs between telogenetic and eogenetic settings have illustrated hydrological response differences (Florea and Vacher 2006, 2007). For example, in eogenetic karst that does not receive allogenic recharge (Almequem 2023), the fraction of diffuse recharge can be larger than the fraction of discrete recharge (Ritorto et al. 2009; Moore et al. 2010; Langston et al. 2012; Brown et al. 2014), which results in pronounced seasonality and longer hydrograph recessions relating to the slower recharge aggregation and longer residence times in the carbonate matrix (Florea and Vacher 2007; Spellman, Gulley, et al. 2022). At the event scale, this behaviour may produce more well mixed, near chemostatic effects, which have been previously observed in primarily diffuse flow karst aquifers (Garrels 1965; Shuster and White 1971; Ternan 1972; Lee and Krothe 2001; Peterson et al. 2002). In contrast, flashy and rapid solute transport is typically observed when spring discharge is largely controlled by discrete autogenic and allogenic recharge through conduits embedded in a low permeability matrix (Lakey and Krothe 1996; Rowden et al. 2001; Peterson et al. 2002; Stueber and Criss 2005; Mahler et al. 2008; Pronk et al. 2009; Yang et al. 2013; Huebsch et al. 2014). Thus, event scale hysteresis effects in eogenetic karst aquifers that contain high matrix storage and variably integrated conduit porosity may show attenuated and delayed solute concentration changes relative to discharge, if any signal is observed at all.

In this study, we examine how high matrix permeability eogenetic karst aquifers respond to recharge events by documenting solute responses and quantifying hysteresis effects at two karst springs in the eogenetic Upper Floridan Aquifer (UFA). The UFA maintains high matrix permeability as well as extensive conduit networks (Florea and Vacher 2007; Gulley et al. 2011; Brown et al. 2019). Recharge to the UFA is primarily diffuse, and discrete recharge at the event scale has not been examined (Ritorto et al. 2009; Langston et al. 2012; Brown et al. 2014; Spellman, Gulley, et al. 2022). We use continuous discharge, nitrate and specific conductance to track discrete recharge, analyse solute behaviour and quantify hysteresis effects which provide insight on the efficiency of integrated porosity (i.e., conduits and fractures) as well as the land uses contributing to storm-induced solute changes. Data from deployable water quality sensors are used because they can capture small or delayed water chemistry changes that may be missed from discrete sampling. Our analysis builds off previous work that documented seasonal and long-term solute variability in the UFA and adds to the conceptual model of residence time distributions in eogenetic karst aquifers (Spellman, Gulley, et al. 2022). Further, because contamination in the UFA from agricultural and urban land use has amplified in recent years, understanding the transport and mobility of

contamination across all timescales is needed locally for comprehensive water resource and land use planning.

1.1 | Site Descriptions

Manatee Spring and Madison Blue Spring are first magnitude (discharge $> 2.85 \text{ m}^3/\text{s}$), phreatic springs that discharge from the UFA (Spellman, Gulley, et al. 2022). Monitoring stations at both springs record discharge, specific conductance and nitrate at 15-min intervals using in situ sensors. Water quality data is quality checked by the Suwannee River Water Management District (SRWMD) in cooperation with the United States Geological Survey (USGS) and is available from 7/2014-Present. Discharge was measured at both stations using stage and adopted rating curves by the USGS in cooperation with the SRWMD and is available for Madison Blue Spring from 1/2001-Present and for Manatee Spring from 4/2002-Present. Specific conductance is measured using a YSI EXO2 conductivity/temperature sensor. Nitrate is measured with a Submersible Ultraviolet Nitrate Analyser (SUNA) by SeaBird Scientific that uses ultraviolet light to estimate nitrate (NO_3^-) as nitrogen ($\text{NO}_3\text{-N}$).

The groundwater contributing areas are uncertain in this karst region, and many boundaries have been drawn with widely varying extents. However, using a combination of dye tracing, groundwater modelling and potentiometric surface maps, a rough estimate of the groundwater contributing area to Manatee Spring is about 80 km^2 (Upchurch et al. 2004), whereas, Madison

Blue Spring likely has a contributing basin of about 259 km^2 (Suwannee River Water Management District (SRWMD) 2004).

Manatee Spring discharges to the lower Suwannee River (Figure 1B) via a spring run that is roughly 300–400 m from the spring vent to the river, and sensor measurements are taken at about $\sim 100 \text{ m}$ downstream from the vent. Madison Blue Spring is located farther north and discharges to the Withlacoochee River (Figure 1C). The spring run is much shorter than the Manatee Spring run (only $\sim 40 \text{ m}$), and the sensor is located just before the confluence of the spring run with the Withlacoochee River. Both springs are impacted by nutrients that are primarily from agricultural (e.g., row crop, livestock) sources but also come from septic sources (Katz 2004).

Manatee Spring and Madison Blue Spring each have distinctive morphological characteristics, but land uses across their groundwater recharge basins are similar. Complex cave hydraulics have been observed, and one sinking stream or sinkhole can be hydraulically linked to multiple spring outlets that can shift under different hydraulic conditions (Greenhalgh et al. 2016a, 2016b), making exact delineations of static groundwater contributing areas difficult. The northern spring, Madison Blue Spring, likely has more groundwater contributing area that is under aquifer confinement based on modelling efforts (Spellman, Gulley, et al. 2022), while the groundwater contributing area to Manatee Spring is primarily unconfined (Gulley et al. 2014). The highest density of agriculture in the SRWMD, specifically row crops, exists where the UFA is unconfined immediately surrounding

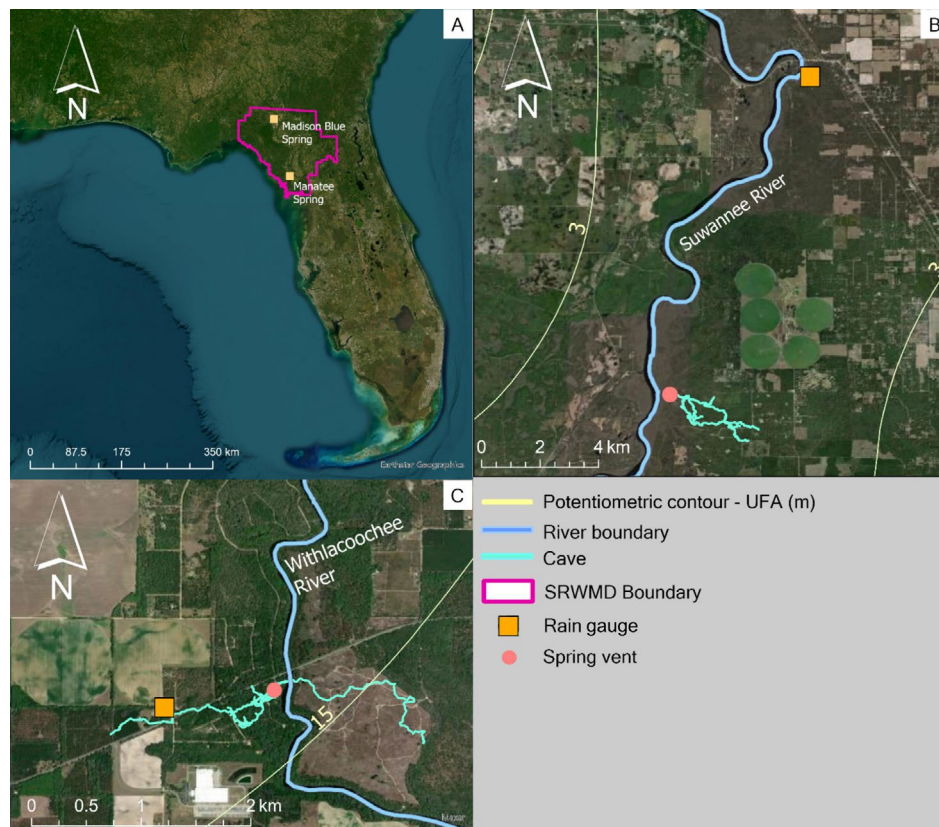


FIGURE 1 | (A) The Suwannee River Water Management District (SRWMD) and the locations of the studied springs within that boundary. (B) The Manatee Spring regional setting with the cave map, rain gauge and spring vent locations. (C) The Madison Blue Spring regional setting with the cave map, rain gauge and spring vent locations.

Madison Blue Spring (Marella et al. 2016). The depth to the water table at Madison Blue Spring is about 12m on average. In contrast, the depth to the water table at Manatee Spring is about 4m, but the water table depth can be highly variable throughout the cave system. Madison Blue Spring has a more complex cave morphology than Manatee Spring with a likely connection to a northern spring system (M2 Blue) (Supporting Information: L1). The possible cave extension from Madison Blue Spring to the northern M2 Blue cave system potentially runs parallel to the Withlacoochee River. It could therefore intersect waters recharged from different land uses, capturing water along the regional hydraulic gradient that would eventually emerge at Madison Blue Spring. However, full exploration of all the tunnels has not been completed, and unknown cave sections likely exist. Both Madison Blue Spring and Manatee Spring are estavelles whereby river water intrudes into the aquifer through the spring vents during high river stages. These 'spring reversals' can last for weeks before all the river water is extruded and only groundwater begins discharging from the vents. Madison Blue Spring reverses more frequently (at least once a year) than Manatee Spring, which can go a few years without a true reversal.

2 | Methods

2.1 | Data Collection and Screening

We downloaded the continuous discharge, specific conductance and nitrate data from the USGS Water Watch website for a time-frame when all data was labelled as approved and quality checked (1/2016–12/2021). Daily precipitation data from two weather stations near each spring (Figure 1B,C) were downloaded for the same dates as the spring data from the SRWMD website (<https://www.mysuwanneeriver.com/>). We also screened the data for periods when they were not representative of groundwater conditions. Specifically, in northern Florida, spring reversals are common at several springs and occur when a river's stage exceeds the hydraulic head in the adjacent aquifer, forcing river water to intrude through the spring vent into the aquifer. When reversals occur, the data recorded by the sensors do not reflect aquifer water and were removed prior to analysis. Sometimes river water will register at the sensor in the spring run, which is downgradient from the vent, and causes sensor noise but may not indicate river water intruded into the aquifer. These spring run oscillations are screened from our analyses as the data are not representative of the groundwater signature. We therefore screened all discharge and water chemistry data that were not representative of groundwater by looking for simultaneous and rapid changes in temperature, decreases in specific conductance and nitrate and a significant reduction in reported discharge. Where these changes occurred, the data were removed between the points when the water quality and discharge data rapidly changed to when they returned to pre-reversal values (Spellman, Gulley, et al. 2022).

2.2 | Signal Filtering and Smoothing

Manatee Spring has tidal oscillations in discharge at the monitoring location because of lower Suwannee River responses to the tides. These tidal oscillations were eliminated from the discharge data using a low pass Godin filter, which removes signals at

periods ≤ 24 h. The Godin filter is the preferred cleaning method used by the National Oceanic and Atmospheric Administration (NOAA) (Pagendam and Percival 2015; Shirahata et al. 2016) and has been effectively used in previous hydrological studies (Zimmer et al. 2019; Spellman, Breithaupt, et al. 2022). Discharge at Madison Blue Spring and the specific conductance and nitrate signatures at both springs were smoothed using a Savitzky–Golay filter to reduce noise (Husic et al. 2023). Erratic fluctuations from boats and swimmers in the water near the sensors can occur, and proper smoothing can improve data precision without distorting the signal and overall interpretation.

2.3 | Event Selection and Quality Control

Events satisfied our quality assurance criteria for inclusion in our analysis if (1) they had concurrent measurements of discharge, specific conductance and nitrate, (2) variations in nitrate concentration were above the accuracy threshold for the nitrate sensor and (3) changes in flow and water quality parameters occurred in response to precipitation. Initial screening for events involved finding peaks in MATLAB using automated methods and manually selecting the events that had complete records of all the criteria. We maximised event selection by terminating the event when the discharge was within 10% of pre-event flow. The 10% criterion was imposed because of the intensity, frequency and duration of reversals at Madison Blue Spring, which limited event availability for analysis. Further, we selected events where nitrate concentration changes were greater than 0.056mg/L. The specified nitrate sensor accuracy is either 0.056mg/L or 10% of the measured nitrate concentration up to 14mg/L. However, because accuracy decreases with increasing nitrate concentration and measured nitrate is low (< 3 mg/L) at both springs, we opted to use the 0.056mg/L threshold for event selection. We also calculated simple statistics using the continuously monitored data. In detail, we generated trend lines using scatterplots between discharge and specific conductance and discharge and nitrate to determine if the solute behaviour followed a pattern that was consistent between larger and smaller events.

2.4 | Discharge–Concentration Indices

Prior to analysis, we normalised all the time series data for direct comparison between events and sites. Normalisation for discharge (Q) (Equation 1) and solute (C) (Equation 2) data was performed by taking each measurement during the event (i) and subtracting the minimum (min) observed value for the entire event, scaled by the difference between the maximum (max) and minimum values.

$$Q_i^{\text{norm}} = \frac{Q_i - Q_{\min}}{Q_{\max} - Q_{\min}} \quad (1)$$

$$C_i^{\text{norm}} = \frac{C_i - C_{\min}}{C_{\max} - C_{\min}} \quad (2)$$

After normalisation, hysteresis curves were generated, then Q – C indices (i.e., Hysteresis Index (HI) and Flushing Index (FI)) were calculated based on the methods proposed by Lloyd et al. (2016). We used modified, normalised HI and FI values

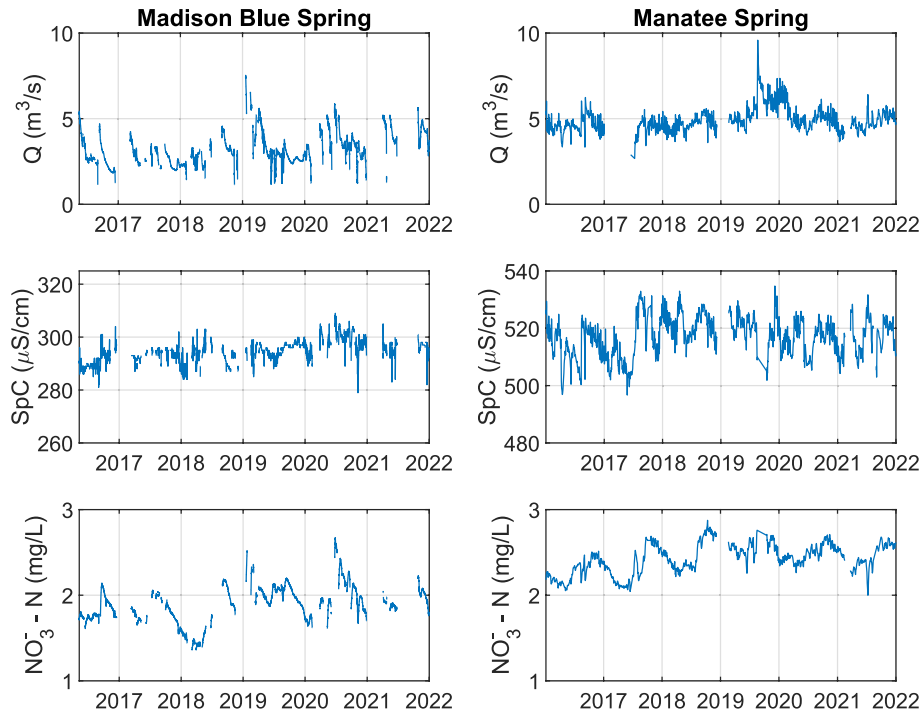


FIGURE 2 | The time series data of discharge (Q), specific conductance (SpC), and nitrate (NO_3-N) for Madison Blue Spring and Manatee Spring. The gaps in the data series represent the effects of filtering and screening for spring reversals. Some sharp changes in discharge remain as discharge will begin to decrease due to river damming, and this occurs before water quality changes are observed.

to define our dataset as they feature less bias, improve interpretations and the normalised values can be used to make comparisons across different sites (Zuecco et al. 2016; Lloyd et al. 2016). The HI is a concise method to capture comparative solute concentration dynamics on the rising limb (RL) and falling limb (FL) of the discharge hydrograph (Equation 3). For the screened data, the RL was defined to be when the discharge hydrograph began to increase until the peak discharge, while the FL was classified as the period from peak discharge until the hydrograph returned to within 10% of the antecedent discharge prior to the RL. For reference, a positive HI is indicative of CW hysteresis behaviour, whereas a negative HI would indicate CCW behaviour. The FI complements the HI and indicates whether dilution or mobilisation is dominant during the event pulse as discharge increases. Values for FI were calculated by taking the solute concentration at the maximum flow rate and subtracting the pre-event solute concentration (Equation 4). A positive difference indicates the solute's mobilisation, while a negative difference indicates its dilution. We report the mean and standard deviation of the HI and FI for each event.

$$HI = n^{-1} \sum_{i=1}^n C_i^{RL} - C_i^{FL} \quad (3)$$

$$FI = C_{Q_{max}} - C_{Q_{min}} \quad (4)$$

3 | Results

The time series data of discharge, specific conductance and nitrate for each spring highlight differences and similarities

between the two systems (Figure 2). During dry weather, Manatee Spring and Madison Blue Spring had average discharges of around 4 and 2 m³/s, respectively, highlighting the significant volume of stored water from the aquifer matrix. The effects of seasonality on flow were also observed in the discharge dataset, particularly at Madison Blue Spring, with higher flow during the summer and lower flow in the winter. Both specific conductance and nitrate have long-term increasing trends based on previous legacy data analysis (Spellman 2021), which can partly be observed in our 4-year assessment period. Nitrate values were similar between the two systems (typically 2–3 mg/L), but Manatee Spring had a more pronounced annual periodicity, likely due to less frequent reversals that can mask aquifer dynamics and the conveyance of seasonal applications of fertiliser (Spellman, Gulley, et al. 2022).

A total of seven events at Madison Blue Spring and 17 events at Manatee Spring met the solute selection criteria for further analysis (Figure 3). The selected events were screened based on (1) if complete discharge, specific conductance and nitrate datasets were available throughout an event, (2) if the changes in nitrate were above the minimum accuracy threshold of the SUNA nitrate sensor (i.e., 0.056 mg/L) and (3) if the parameter variations were in response to a precipitation event. At Madison Blue Spring, many events that were large enough to generate substantial flow also initiated spring flow reversals, resulting in only eight qualifying events. However, the nitrate threshold (Criterion 2) resulted in the removal of another event, bringing the final number of events to seven. Two events (Events 1 and 5) were close to, but did not quite return to, 10% of the pre-event flow. They were included in the analysis, though, to capture as many hysteresis events as possible.

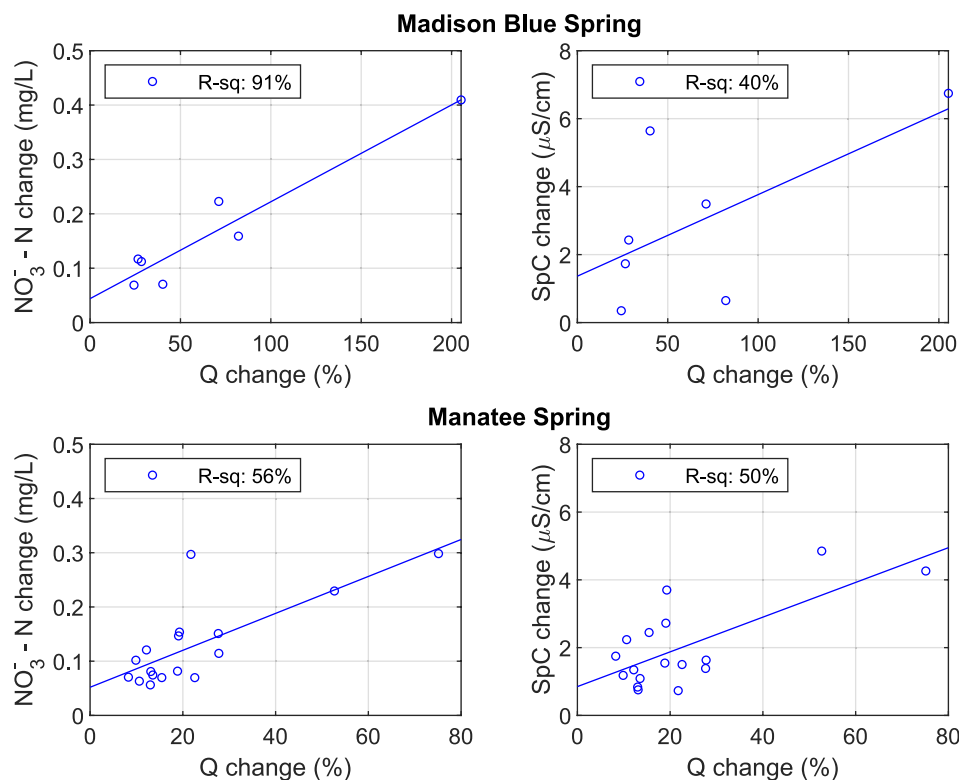


FIGURE 3 | The relationship between the absolute change in nitrate ($\text{NO}_3^- - \text{N}$) and the percent change in discharge (Q) as well as the absolute change in specific conductance (SpC) and the percent change in discharge (Q) for both spring sites.

For Manatee Spring, 25 events met Criteria 1 and 3, but only 17 of those 25 also met the nitrate concentration criterion.

The scatterplots and trend lines for the changes in specific conductance and nitrate across the selected events against percent change in discharge (Figure 3) illustrate relatively linear responses. The relationship strength (as R^2) for specific conductance was strongest at Manatee Spring, while Madison Blue Spring had the strongest relationship between percent change in discharge and the difference in nitrate concentrations. Overall, changes in specific conductance and nitrate were small during events even when the percent change in discharge was large. The change in discharge was also consistently higher at Madison Blue Spring than at Manatee Spring. Similar patterns in solute behaviour across events suggest that even small events are useful for hysteresis interpretations regardless of their relatively minor solute variations.

The normalised discharge and concentration values for each event further demonstrate their observed variations in the springs (Figure 4; note that the unadjusted data are in Table 1 and Figure 2). At both springs, most solute minima or maxima occurred after peak discharge, but a few events deviated from this behaviour. At Manatee Spring, Events 3, 9, 11 and 14 featured solute minima or maxima prior to peak discharge. Event 3 was the only mobilisation event observed for nitrate, which peaked before discharge. During Event 9, nitrate reached a minimum before discharge peaked, while Events 11 and 14 showed specific conductance peaking slightly before discharge. Only two events at Madison Blue Spring, Events 2 and 6, showed dilution of nitrate. Event 6 at Madison Blue Spring also showed an initial dilution in specific conductance, and both parameters

increased around peak discharge, remaining elevated for the rest of the event window. We found that, in general, (1) storm event responses at Madison Blue Spring were much longer (10 ± 8 days) than those at Manatee Spring (3 ± 2 days) and were slower to reach peak discharge, (2) specific conductance generally increased as storm responses progressed and (3) nitrate had contrasting responses at the two springs, with concentrations typically increasing as events progressed at Madison Blue Spring but decreasing with event progression at Manatee Spring.

The combination of solute mobilisation or dilution and the timing of the solute minima or maxima with respect to discharge is reflected in the hysteresis curves (Figures 5 and 6). Manatee Spring showed mostly CCW behaviour for specific conductance, except for Events 11 and 14, where specific conductance peaked before discharge. Thus, the hysteresis behaviours were CW in these cases. For nitrate at Manatee Spring, hysteresis was dominated by CW behaviour, except for Event 9, which was CCW. The CCW behaviour at Event 9 is likely due to the low initial concentration of nitrate prior to hysteresis. At Madison Blue Spring, hysteresis for specific conductance and nitrate was almost all CCW, except for Events 2 and 7 for specific conductance and Event 2 for nitrate. A few hysteresis curves displayed complex, Figure-8 behaviour. Figure-8 behaviour was observed at Manatee Spring for specific conductance during Events 13, 16 and 17 and for nitrate during Events 8, 16 and 17. Madison Blue Spring showed Figure-8 behaviour in specific conductance during Events 1 and 5 and in nitrate only for Event 5.

The HI and FI indices illustrate the storm response behaviour for each site (Figure 7). We follow the interpretation of Waite

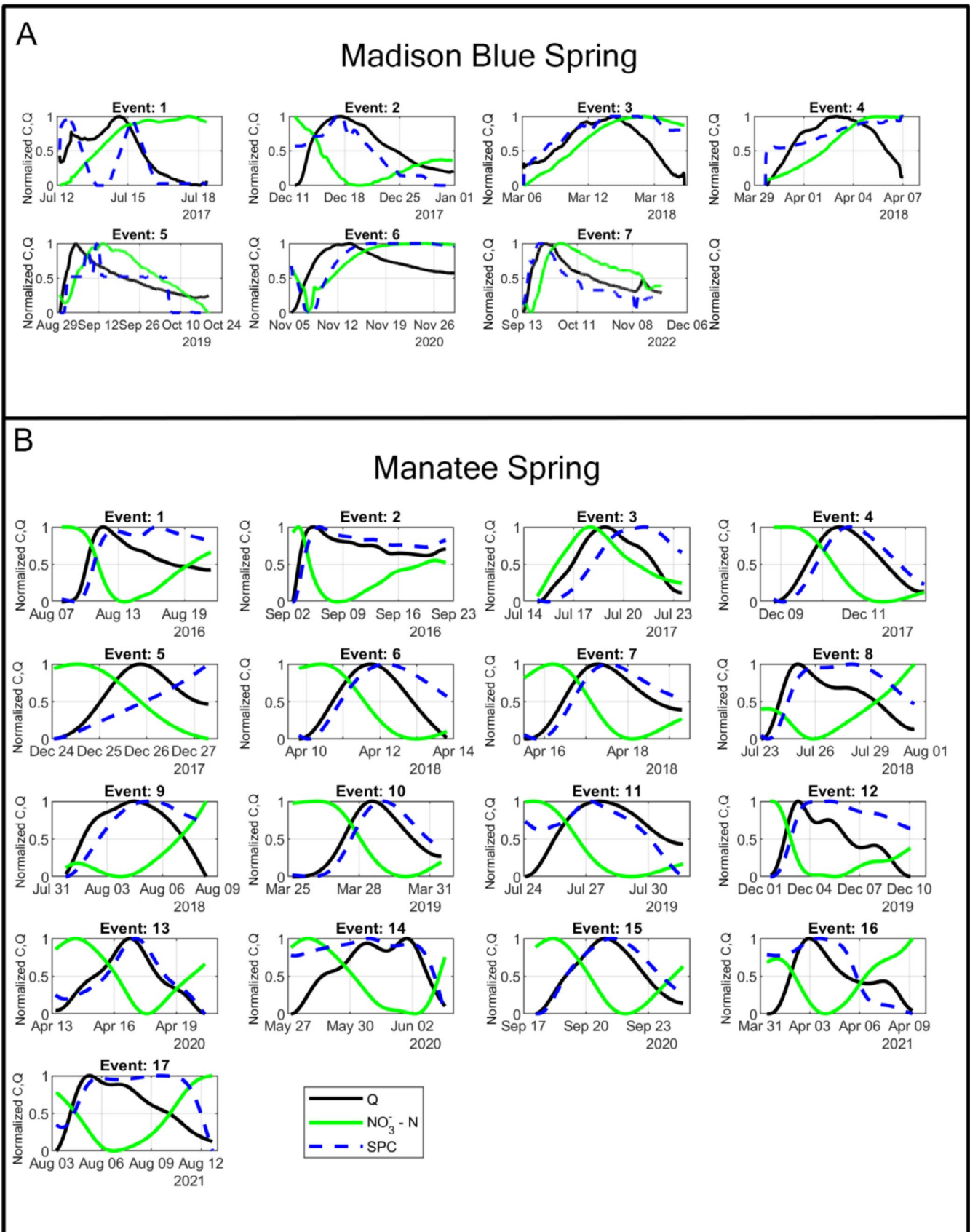


FIGURE 4 | The selected discharge (Q) events at Madison Blue Spring and Manatee Spring for the hysteresis analysis. The plotted discharge (Q), nitrate ($\text{NO}_3\text{-N}$) and specific conductance (SpC) data are normalised to event minima and maxima and thus range from 0 to 1.

TABLE 1 | Rainfall, discharge, temperature and solute concentration data.

Manatee Spring												
Event	<i>P</i> (mm)	<i>T</i> _{min} (°C)	<i>T</i> _{max} (°C)	<i>T</i> _{initial} (°C)	SpC _{initial} (μS/cm)	SpC at peak change (μS/cm)	NO ₃ -N initial (mg/L)	NO ₃ -N at peak change (mg/L)	<i>t</i> _{PEAK} (days)	<i>t</i> _{DUR} (days)	<i>Q</i> _{INITIAL} (m ³ /s)	<i>Q</i> _{MAX} (m ³ /s)
1	67.6	22.4	22.6	22.4	501.20	524.60	2.37	2.14	1.5	5.4	4.0	6.0
2	100.8	22.4	22.6	22.6	502.16	523.53	2.45	2.17	1.1	7.8	3.6	6.2
3	14.7	22.5	22.7	22.5	515.39	527.65	2.35	2.45	1.6	3.5	3.5	4.1
4	50.8	22.4	22.5	22.5	511.32	518.93	2.57	2.49	0.6	1.5	4.4	5.3
5	0.5	22.4	22.5	22.4	525.44	530.87	2.31	2.24	0.7	1.3	4.5	5.1
6	58.7	22.4	22.6	22.4	516.36	522.95	2.32	2.27	0.8	1.6	4.6	5.2
7	17.8	22.4	22.6	22.6	519.13	525.26	2.40	2.39	0.7	1.5	4.3	5.2
8	51.6	22.5	22.7	22.5	515.79	519.59	2.61	2.53	0.8	3.3	4.8	5.3
9	14.5	22.5	22.7	22.6	525.23	526.25	2.67	2.38	1.5	3.1	4.9	5.3
10	15.0	22.4	22.6	22.5	522.19	524.20	2.39	2.33	1.4	2.5	4.8	5.4
11	36.3	22.5	22.6	22.6	515.06	522.16	2.59	2.46	0.7	1.3	5.2	6.3
12	1.3	22.5	22.7	22.7	525.92	528.38	2.26	2.22	1.4	2.9	5.6	6.6
13	22.1	22.6	22.8	22.7	511.84	520.84	2.37	2.26	1.1	0.5	4.6	5.7
14	8.4	22.7	22.8	22.7	521.14	525.50	2.56	2.50	1.6	0.5	4.8	5.2
15	11.4	22.6	22.7	22.7	525.52	532.96	2.29	2.23	0.7	0.5	4.8	6.1
16	0.3	22.5	22.8	22.8	515.68	534.75	2.67	2.51	0.7	0.6	4.7	5.2
17	49.5	22.6	22.9	22.7	519.54	525.87	2.36	2.26	0.8	0.6	4.6	5.5
Madison Blue Spring												
Event	<i>P</i> (mm)	<i>T</i> _{min} (°C)	<i>T</i> _{max} (°C)	<i>T</i> _{initial} (°C)	SpC _{initial} (μS/cm)	SpC at peak change (μS/cm)	NO ₃ -N initial (mg/L)	NO ₃ -N at peak change (mg/L)	<i>t</i> _{PEAK} (days)	<i>t</i> _{DUR} (days)	<i>Q</i> _{INITIAL} (m ³ /s)	<i>Q</i> _{MAX} (m ³ /s)
1	0.8	21.2	21.6	21.3	299.58	310.03	1.81	2.04	0.69	23.2	2.7	4.8
2	16.3	20.6	21.0	20.9	279.25	295.00	1.43	1.36	0.41	3.3	4.8	5.6
3	8.9	20.9	21.0	20.9	289.00	296.02	1.56	1.67	0.39	8.0	2.0	2.9
4	0.3	20.8	21.0	20.9	291.95	297.00	1.37	1.48	0.48	5.8	2.1	2.7
5	27.2	20.9	21.1	20.9	294.97	296.00	1.98	2.05	0.49	2.7	2.7	3.3
6	26.9	21.1	21.2	21.1	296.00	297.92	2.04	2.20	0.55	20.2	2.9	3.5
7	11.4	20.6	21.2	21.2	279.17	298.00	1.57	1.98	0.54	9.5	1.9	3.8

Note: Bolded solute concentrations indicate dilution occurred.

et al. (2023) with respect to the combination of solute dilution or mobilisation (FI) and CW or CCW hysteresis (HI), using those behaviours to interpret rapid or delayed response timing for the spring systems. In general, positive HI values were associated with Manatee Spring's nitrate hysteresis, whereas, negative HI values were associated with nitrate hysteresis at Madison Blue Spring. Both springs usually featured negative HI values for specific conductance hysteresis. Positive HI values for specific conductance were calculated for two events at Manatee Spring (Events 11 and 14), which also showed specific conductance peaks prior to the discharge peaks. At Madison Blue Spring, three events (Events 1, 2 and 5) had positive HI values

for specific conductance. Two of those events had Figure-8 behaviour (Events 1 and 5), while Event 2 had CW behaviour. Manatee Spring more consistently demonstrated dilution and CW behaviour for nitrate, resulting in a negative HI for only two events (Events 9 and 14). Overall, most events (17 out of 24) displayed delayed connectivity, while the remaining events (seven out of 24) plotted as having more rapid connectivity (Figure 7). The FI values (Figure 7) were generally reflective of the nitrate mobilisation at Madison Blue Spring and nitrate dilution at Manatee Spring, as well as the specific conductance mobilisation at both springs, as reported previously (Figure 4), but a few events were exceptions.

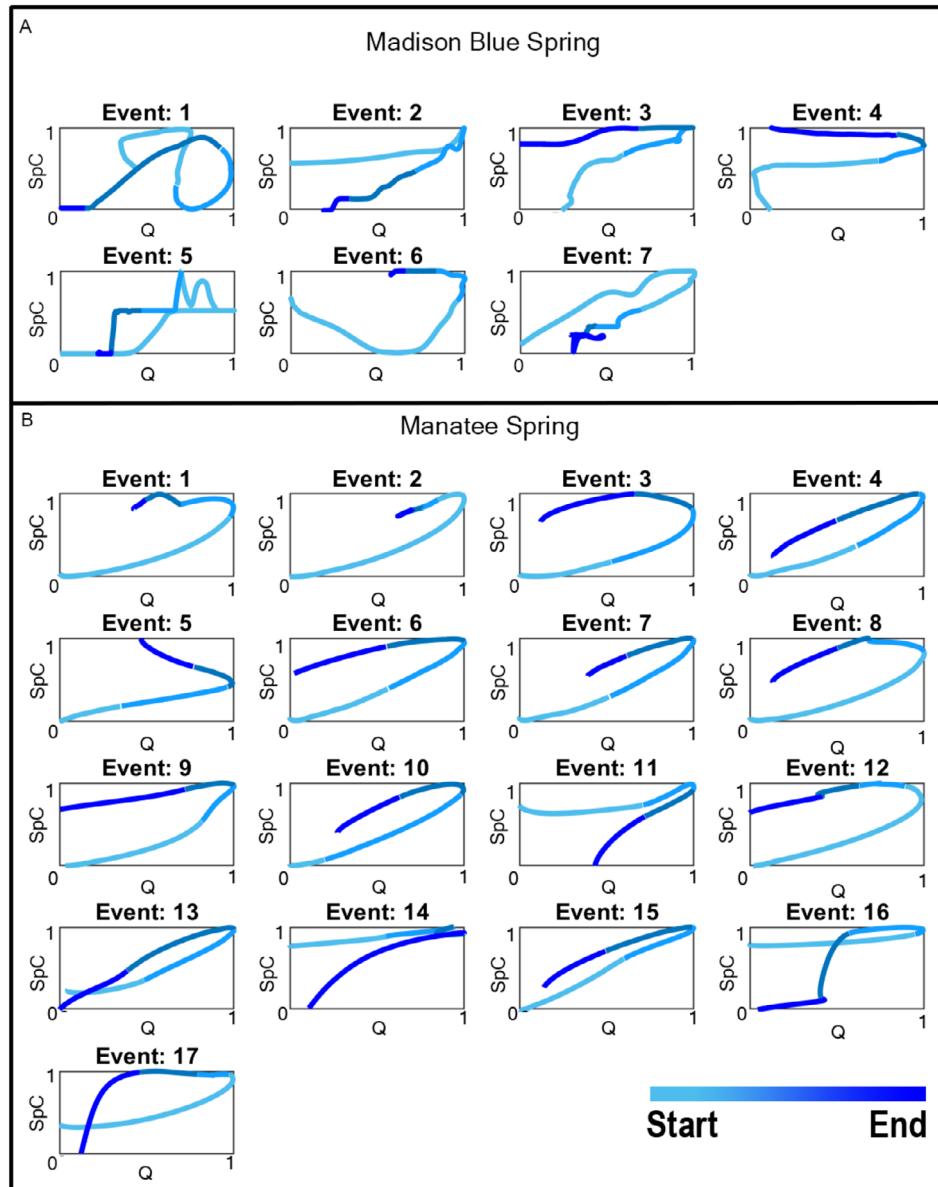


FIGURE 5 | Madison Blue Spring's and Manatee Spring's normalised specific conductance (SpC) and discharge (Q) hysteresis responses for all the selected events.

4 | Discussion

We presented evidence of an attenuated quickflow response to storms in an eogenetic karst aquifer and quantified specific conductance and nitrate hysteresis to holistically characterise solute behaviour over multiple storm events. We identified similarities and differences between the two springs regarding specific conductance and nitrate response, which guide the interpretation of the potential causes for these patterns. Hereafter, we summarise previous investigations of event-based specific conductance and nitrate behaviour in other karst aquifers and use the results to discuss the causes of combined solute behaviour. We then examine what the hysteresis metrics, HI and FI, can tell us about aquifer hydrodynamics and how our results compare to other studies. We finally discuss the integration of our results into local water resource management.

4.1 | Solute Responses and Causes

We begin by summarising the results of other karst studies that documented specific conductance and nitrate changes in karst springs during and after storm events using newly developed conceptual figures (Figure 8). Using Figure 8, we describe different phases (Phases 1, 2, 3 and 4) that are segmented in each graph and numbered. Phase 1 is the chemical baseline condition that continues until just after the discharge increases and the initial water stored in the conduit has been displaced (Dreiss 1989). After the initial conduit water is displaced, most classic karst systems show dilution of solutes (Phase 2) as rapid recharge through sinking streams dilutes the concentrations relative to the baseline aquifer water (Ryan and Meiman 1996; Lakey and Krothe 1996; Vesper and White 2004; Birk et al. 2004; Mahler et al. 2008; Mahler and Garner 2009; Berglund et al. 2019; Husic

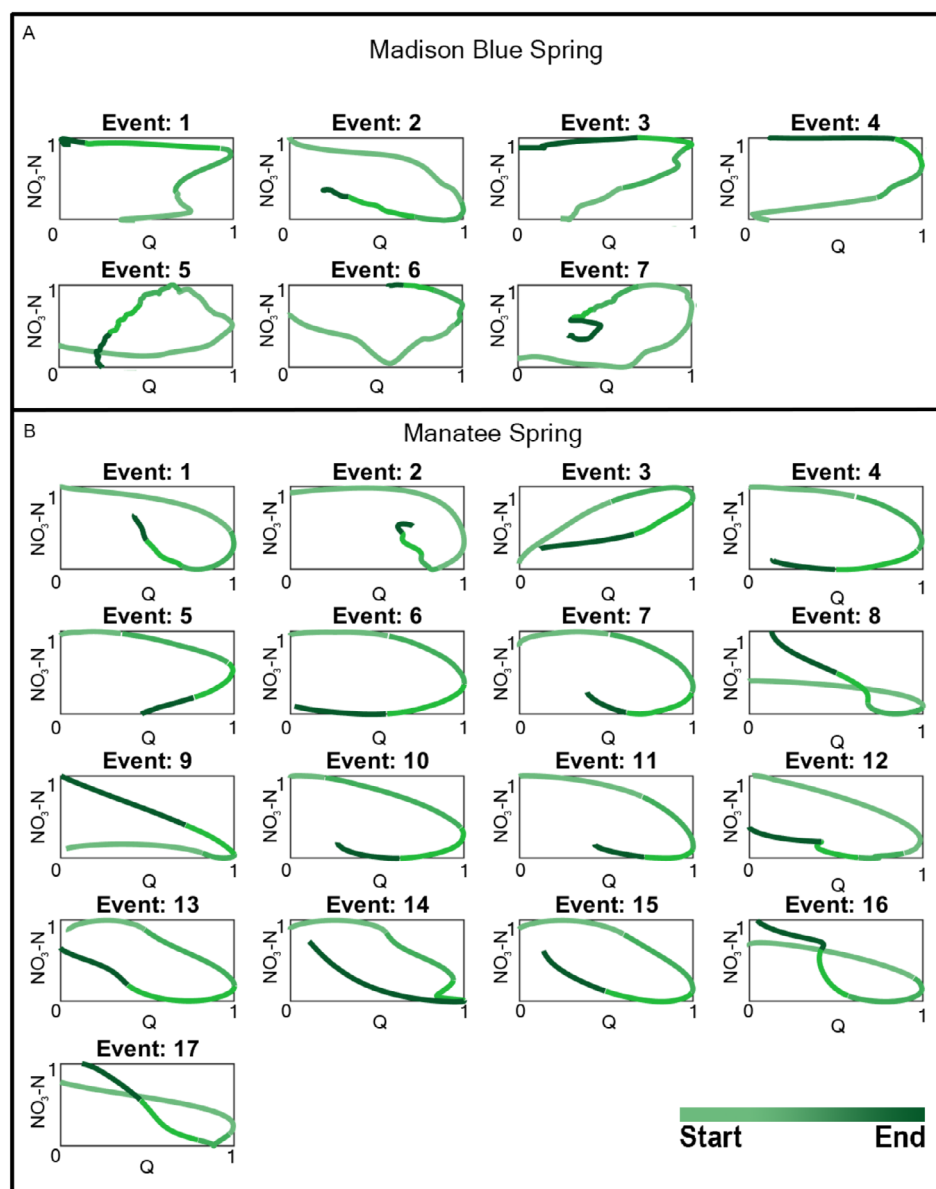


FIGURE 6 | Madison Blue Spring's and Manatee Spring's normalised nitrate ($\text{NO}_3\text{-N}$) and discharge (Q) hysteresis responses for all the selected events.

et al. 2023). However, some studies analysing nitrate have shown its mobilisation if runoff to the sinking stream is enriched with nutrients, which can occur after intensive fertilisation (Rowden et al. 2001; Pronk et al. 2009; Huebsch et al. 2014). Furthermore, increases in specific conductance have been observed when the application of road salts (Toran and Reisch 2013; Robinson and Hasenmueller 2017) or other anthropogenic alterations to the landscape and groundwater (Doctor et al. 2006) have occurred. In some studies, usually under dry conditions (Toran and Reisch 2013), a flushing of mineralised older water from the aquifer matrix may occur, which commonly appears before peak discharge (Tagne and Dowling 2018; Berglund et al. 2019), but can occur after peak discharge (Phase 3) in certain cases (Desmarais and Rojstaczer 2002). In Phase 4, solutes will return to pre-event conditions, unless an anthropogenic source (e.g., road salts or agricultural amendments) keeps the solute concentrations elevated (Rowden et al. 2001; Toran and Reisch 2013). Specific conductance has also been shown to remain elevated

for several days after storm events due to shifting catchment boundaries (Ravbar et al. 2011).

When coupling our solute time series data with the results of previous studies (Figure 8), we can infer the general mechanisms driving the solute behaviours. The time series data of specific conductance at both springs as well as nitrate at Madison Blue Spring align with literature results in Figure 8D and sometimes Figure 8E, whereas, the nitrate patterns at Manatee Spring align with literature results in Figure 8A and sometimes Figure 8B. The previous studies that documented these types of responses concluded the solute changes were either from discrete recharge or older water being evacuated from storage (see Figure 8D). Between those two causes, solute changes in both springs studied herein are likely from new rainwater moving through fractures or discrete features that are integrated into conduits contributing to the springs' discharges. Cave divers have observed vertical fractures in the phreatic caves, where sediment

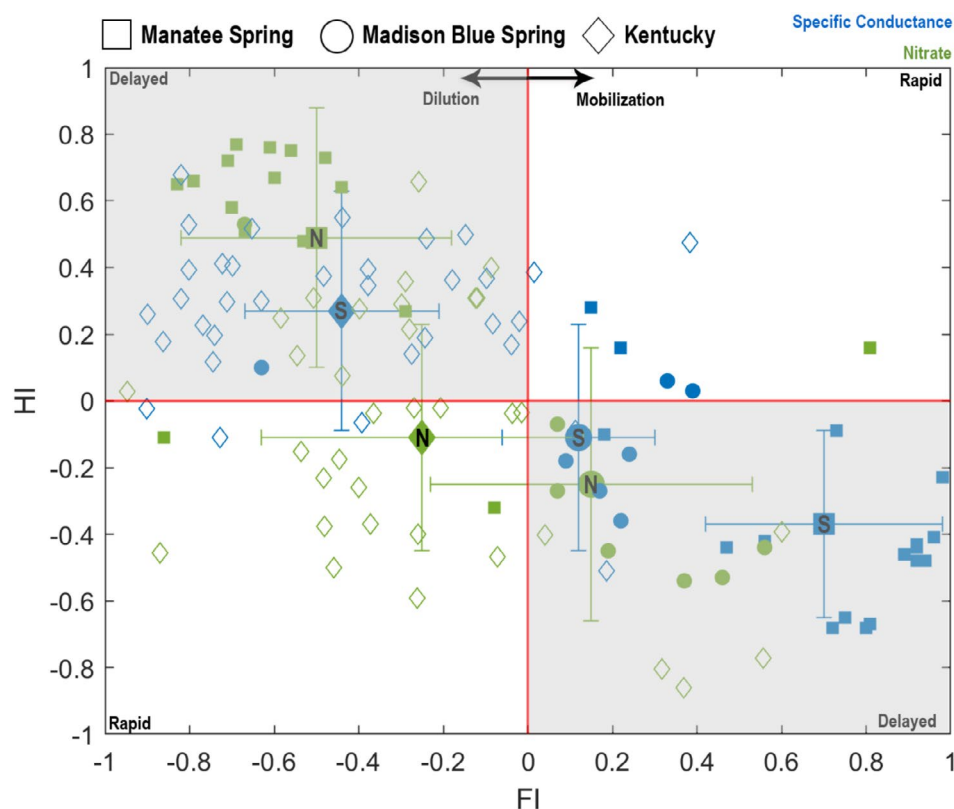


FIGURE 7 | A scatterplot of the average (large shapes) and individual (small shapes) HI and FI values calculated from the selected events at Madison Blue Spring (circles) and Manatee Spring (squares). Green shapes represent nitrate (N) responses, while blue shapes represent specific conductance (S) responses. The bars on the average values show the standard deviation within the individual datasets. Red lines are the zero markers for each index and separate the graph into four quadrants that represent characteristic hysteresis behaviour. The telogenetic karst spring in Kentucky is shown with diamonds and individual sites are not filled. The interpretation of connectivity (rapid, delayed) and its relationship to dilution and mobilisation is taken from Waite et al. (2023). All data used to create this figure are available in [Supporting Information: E1 and E2](#).

has infiltrated and cooler temperatures can be felt, providing evidence for water pathways integrated into the main conduit, and some of these fractures are included in some cave maps. The increases in specific conductance could be from dissolved constituents in the overlying soil or recharge near farms. For example, specific conductance data collected from shallow wells immediately downgradient from a major farming operation near Manatee Spring have values above $1000\mu\text{S}/\text{cm}$ (Spellman et al. 2023; Table S1). Nevertheless, previous literature attributed increases in specific conductance to mineralised water being mobilised in the vadose zone upon infiltration (Figure 8D; Desmarais and Rojstaczer 2002). Matrix water is also likely to have elevated specific conductance, providing another potential source of high specific conductance water to the springs. Because the caves are entirely phreatic and have conduit depths of $>8\text{m}$ (Supporting Information: L1), the likelihood of mineralised water in the vadose zone moving entirely through the diffuse matrix to the conduit is unlikely on the timescales under which the solute increases were observed (Figure 4). However, diffuse water from the matrix could be mobilised into vertical fractures that cut through the vadose zone and are integrated with the conduit network, potentially explaining the observed responses.

The causes of the nitrate and specific conductance responses would be similar for each spring. In the case of Madison Blue

Spring, a conduit has recently been mapped very close to a nearby farm (Figure 1C) and could be the source of nitrate to the spring. The spatially extensive cave system at Madison Blue Spring, which is within a high-density agricultural region, could also intersect more agricultural land uses than the Manatee Spring system. Intense agricultural activities in the recharge area of Madison Blue Spring could thus be a continued source of solutes, explaining the longer duration of the elevated nitrate and sometimes specific conductance signatures on the FLs of the hydrographs for the selected events (Figure 3), which behave similarly to Figure 8E (Birk et al. 2004; Reisch 2010; Huebsch et al. 2014). For Manatee Spring, the dilution of nitrate could be explained by drainage into a nearby sinkhole from natural landscapes or from non-fertilised crops such as peanuts, which are common in the area. Unfortunately, we could not meaningfully interpret seasonal influences on hysteresis dynamics due to the lack of representative events from different seasons at the two sites. Furthermore, the cause of the complex (e.g., Figure-8) solute responses could be from additional recharge through other discrete features that may reach the spring vents on different timescales or could be due to shifting hydraulic boundaries (Raybar et al. 2011; Husic et al. 2023). Without additional geochemical data, the reasons for complex Figure-8 responses are difficult to confirm. More detailed geochemical analyses at the event scale, such as major ions, calcium/magnesium ratio

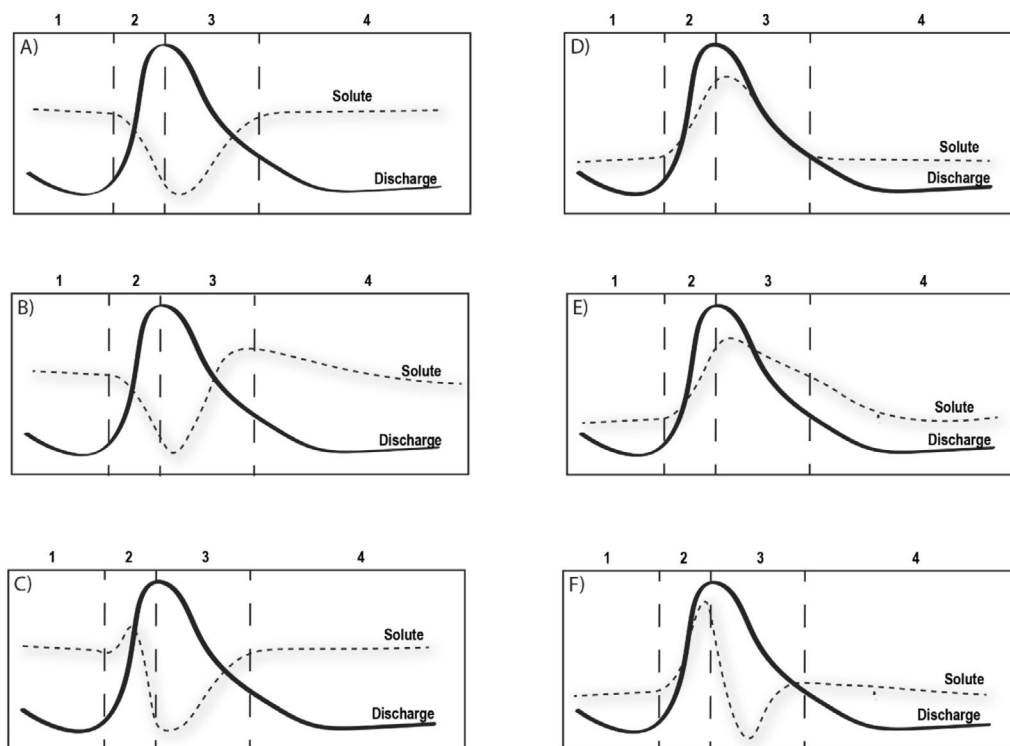


FIGURE 8 | The idealised event-based solute responses recreated and summarised from the literature using the sources referenced in the figures. Also included are the causes of the observed behaviour from each study. The numbers at the top of the graph represent the different phases of the responses as discussed in the text. (A) Dilution effect common from sinkholes or swallets. A return to initial concentrations can occur when discharge returns to initial conditions (Tagne and Dowling 2018) or dilution may continue after discharge returns to initial conditions (Vesper and White 2004). (B) Dilution response followed by an increase in solute concentration above initial conditions. This could be from a continuous diffuse source being delivered to conduit (Toran and Reisch 2013) or changing catchment boundaries (Ravbar et al. 2011). (C) An initial increase in solute concentration due to more equilibrated water being displaced by the initial pressure propagation from recharge (Berglund et al. 2019). Dilution may occur soon after as stormwater moves into the conduit. Values return to initial concentrations after discharge goes back to initial conditions. (D) Mobilization due to abundant solute supply in swallets or sinkholes into a phreatic conduit (Doctor et al. 2006), or from older aquifer water (Desmarais and Rojstaczer 2002). (E) Similar mobilization to 8D from abundant surface supply (Huebsch et al. 2014). Elevated concentration on falling limb is due to continued diffusely fed supply. This likely occurs under conditions 8D, but with an additional diffuse component that is recharging the conduit. (F) A compound event with mobilization from a surface source and then dilution from another surface source. Likely occurs where multiple discrete sources may deliver recharge from different land uses via swallets, fractures or sinkholes (Rowden et al. 2001).

and water isotope data, would provide additional insights into these complex responses as well as the causes of specific conductance changes.

We note the solute changes at our selected sites were small relative to those detected in other studies, although the variations in specific conductance were similar in magnitude to Ravbar et al. (2011) and Opsahl et al. (2017). The attenuation of the solute response is likely due to recharge through a small number of discrete features, whereby the recharged water is mixed with the large volume of conduit water that is continuously sustained via the carbonate matrix. This process likely affected the number of events that met our nitrate criteria threshold. One event at Madison Blue Spring and eight events at Manatee Spring were not above 0.056 mg/L, although changes to nitrate concentrations were still observed (Supporting Information: E1). These small nitrate concentration changes would likely be due to the dilution of recharged water with stored groundwater, whereby not enough solute flux entered the aquifer to cause the changes required to meet our criteria. Though events that did not meet our nitrate criteria threshold were omitted to ensure

interpretation was justified, the fact that some changes were even smaller than the threshold shows how substantial attenuated discrete recharge signals are in the UFA, which could be missed with discrete sampling.

The hysteretic behaviours of specific conductance and nitrate that we observed at our study sites capture the average behaviours of each spring basin and are consistent with other studies in similar phreatic cave systems. Previous work in karst aquifers where caves were either fully or partially saturated demonstrated distinct hysteresis patterns between the two hydraulic conditions (Bettel et al. 2022). When the aquifer was 'full' and caves were entirely phreatic, CCW hysteresis patterns were observed. In contrast, when the caves were partially saturated, CW hysteresis patterns occurred. These responses are consistent with the delayed delivery of the solutes to springs in karst (Schilling et al. 2019; Toran and Reisch 2013) and other even river settings (Malutta et al. 2020; Liu et al. 2021). Researchers made similar interpretations at Royal Spring in Kentucky (United States), where CW behaviour was observed when sinkholes dominated recharge

volumes, thereby diluting the spring's nitrate concentrations (Husic et al. 2023). In contrast, CCW behaviour was observed when a hydrodynamic forcing (i.e., recharge) generated connectivity between the epikarst and phreatic zone. In this situation, nitrate-rich water entered the conduit system at the latter portion of the discharge hydrograph. This second process that produces CCW behaviour would be similar to the pathways required for recharge to enter the UFA phreatic cave network, where recharge waters would need to enter via fractures or sinkholes that are hydraulically integrated with the conduits. In general, hysteresis shows that porosity (fracture and conduit) integration with the land surface is present, but diffuse recharge and mixing with the existing stored groundwater substantially impact the hysteresis response.

To put our results into a broader context, we compared the hysteresis behaviour captured by the HI and FI values for springs in the eogenetic UFA system with responses reported by Husic et al. (2023) for a telogenetic system (Royal Spring) in Kentucky and observed noticeable differences (Figure 7). Royal Spring receives primarily autogenic, discrete recharge through swallets (Husic et al. 2020). The HI versus FI data from the UFA spring sites primarily plot within the top-left ($FI < 0$ and $HI > 0$) and bottom-right ($FI > 0$ and $HI < 0$) quadrants of the graph and suggest delayed transport from the solute source. In contrast, much of the data from the telogenetic setting are plotted in the bottom-left quadrant ($FI < 0$ and $HI < 0$), indicating dilution and efficient connectivity with the surface, although the transport timing varied between rapid and delayed. For example, responses in the direction of hysteresis for nitrate were evenly spread with roughly 1/3 CW, 1/3 CCW and 1/3 Figure-8 behaviour, resulting from the timing differences in nitrate delivery to the spring relative to peak discharge (Husic et al. 2023). However, in the UFA, the behaviour was relatively consistent between events for all solutes with respect to the direction of hysteresis and the interpretation of those hysteresis patterns indicating delayed responses. Additionally, while the solutes from the Kentucky site were primarily diluted ($FI < 0$), mobilisation was the most common response for the UFA springs except for the nitrate at Manatee Spring. Though the telogenetic Kentucky site is in an agricultural area (where runoff can be high in nitrate) much like the spring studied herein, the substantial volumes of water recharged from sinking streams likely caused the observed nitrate dilution. Similarities also occur; however, conflicting solute responses between nitrate and specific conductance were observed in the telogenetic setting as with Manatee Spring herein, however, this is simply a function of source which was discussed previously. Thus, the hydrological differences between telogenetic and eogenetic karst appear to be captured in their hysteretic behaviours, although some overlap occurs. Given the functional differences between these types of karst systems, this result is somewhat expected, although we quantify these differences. It is therefore encouraged that additional hysteresis across more hydrodynamically varied systems be carried out to more accurately capture hysteresis behaviours and make definitive conclusions.

Our findings have implications for water resource evaluations and land management across eogenetic karst aquifers and specifically in the UFA. The largely analogous behaviour of specific

conductance and nitrate for UFA springs during events suggests that primary, discrete recharge areas contributing to spring discharge may be relatively stable and consistently activated. The limited number of events at Madison Blue Spring that met our criteria precludes a definitive conclusion on the stability of the contributing recharge area, however. Further, the linear relationships between changes in discharge and solute concentrations show how precipitation influences the magnitude of solute variability, which is likely the result of the volume of recharged water as well as the availability of the solutes at the land surface. These consequences would be less impactful at an individual spring but are more consequential for storm-induced solute mobilisation across multiple springs in the UFA. For example, even if the change in nitrate is small at a single spring, the total nitrate flux that can be mobilised from regional springs to rivers (Brown et al. 2014) after a storm event could have impacts on nitrate delivery to the Gulf of Mexico. Although continuous loading is more impactful (Zhang et al. 2020) than the event-based mobility shown herein, the solute pulses following storm events still increase their overall fluxes. These effects may become more problematic with changing land use and climate. Future studies should focus on employing more detailed geochemical analyses (e.g., precipitation and spring water isotope data) to identify quickflow signals and quantify the volumes of discrete recharge.

5 | Conclusions

We used high-resolution discharge, nitrate ($\text{NO}_3\text{-N}$) and specific conductance to concisely and holistically characterise hysteresis behaviour and overall solute dynamics in an eogenetic karst aquifer. Our results informed the hydrodynamics of the UFA, including that delayed, discrete recharge emerges at the spring vents, though the changes in solute fluxes are relatively small. The small changes observed are likely due to the dominance of diffuse recharge as well as limited discrete features recharging the aquifer that impact the springs on the timescales that we assessed. Furthermore, the solute responses at each spring are relatively consistent among recharge events, illustrating consistent hydraulic connection with the land surface and discrete recharge features. Our results also illustrate how hysteresis effects in eogenetic karst aquifers show significant differences to telogenetic systems, whereby in telogenetic karst, dilution of solutes and more varied nitrate responses between storms were observed. More analyses at different sites that can incorporate our results would provide further insight.

Data Availability Statement

The data shown in Figure 7 are available in the [Supporting Information](#). All original data can be downloaded from the United States Geological Survey (USGS) website <http://www.waterwatch.usgs.gov> and the Suwannee River Water Management District (SRWMD) website <https://mysuwanneeriver.com>. Additional MATLAB code used to analyse and visualise the data can be made available upon request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.